

Customer Segmentation & Value Analysis

Unlocking business growth through data-driven customer understanding.

Defining the Business Challenge

Business Problem

Retail customer analytics often lacks granular insights into customer behaviour, leading to generalised marketing strategies and missed opportunities for targeted engagement.

Project Objectives

- Identify distinct customer segments.
- Analyse value drivers for each segment.
- Develop actionable insights for personalised strategies.

Dataset Overview & Tooling

Dataset Snapshot

- **Source:** E-commerce Transactional Data
- **Size:** Over 500,000 transactions, 250,000 customers
- **Key Columns:** Customer ID, Purchase Date, Product Details, Price, Quantity

Tools & Technologies



Python

Data manipulation, feature engineering, clustering



SQL & PostgreSQL

Data extraction, querying, initial analysis



Power BI

Interactive dashboards, visualisations

Data Preparation & Enhancement

01

Data Cleaning

- Handling missing values (e.g., imputation for missing customer details).
- Removing duplicate records and inconsistent entries.
- Outlier detection and treatment (e.g., extreme transaction values).

02

Feature Engineering

- RFM (Recency, Frequency, Monetary) values calculation.
- Customer lifetime value (CLTV) projection.
- Categorical encoding for product attributes.

03

Data Transformation

- Normalisation of numerical features for clustering algorithms.
- Aggregation of transactional data to customer level.

Exploratory Data Analysis Highlights

Transaction Trends

Identified seasonal peaks in purchasing behaviour and product categories with highest sales volumes.

Customer Demographics

Analysed distribution of customer locations and average order values by region.

Product Performance

Uncovered top-selling products and those with high return rates, informing inventory decisions.

SQL-Based Business Insights

Revenue Analysis

Quarterly and annual revenue growth trends, identifying periods of high and low performance.

Metric	Q1	Q2
Revenue (£)	2.1M	2.5M
Growth (%)	+10%	+19%

Customer Segmentation

Initial segment identification based on purchase frequency and average spend.

- **High-Value:** Top 10% by spend.
- **Frequent Shoppers:** >5 purchases annually.
- **New Customers:** First 30 days post-acquisition.

Refining Customer Segmentation Logic

1

New Customers

Customers who have made their first purchase within the last 30 days. Focus on onboarding and initial engagement.

2

Returning Customers

Customers with multiple purchases but not yet meeting loyal criteria. Strategies for retention and increased frequency.

3

Loyal Customers

Customers with high RFM scores, consistent purchases over time. Prioritise loyalty programmes and exclusive offers.

Key Findings & Insights

→ High-Value Segment Drive 60% Revenue

A small percentage of loyal customers are responsible for the majority of sales, necessitating tailored retention efforts.

→ Seasonal Product Affinity

Identified strong correlation between certain product categories and seasonal purchase patterns, optimising inventory.

→ Targeted Promotions Boost Engagement

Specific campaigns for 'Returning Customers' resulted in a 15% increase in purchase frequency.

→ Regional Purchase Disparities

Identified geographic variations in product preferences and spending habits, informing localised marketing.

CHAPTER 8

Power BI Dashboard Overview

The interactive Power BI dashboard provides a comprehensive view of customer segments, key metrics, and actionable insights.

Placeholder for screenshot of Power BI dashboard showing customer segments, revenue by segment, and key performance indicators.

Business Recommendations & Conclusion

Strategic Recommendations

- **Personalised Marketing:** Tailor campaigns based on identified customer segments.
- **Loyalty Programmes:** Enhance offerings for high-value and loyal customers.
- **Product Optimisation:** Adjust inventory and promotions based on seasonal and regional insights.
- **Customer Journey Mapping:** Focus on improving experience for new and returning customers.

Impact & Next Steps

Implementing these recommendations is projected to increase customer retention by 10% and overall revenue by 5-7% within the next year.

Thank you!

Questions & Discussion